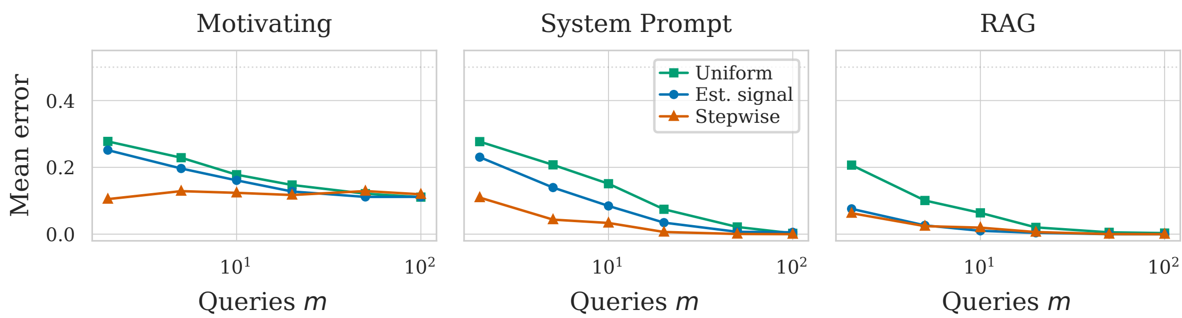

Figure 1. Mean classification error vs. query budget m for three selection strategies across three tasks (n=80). Stepwise selection matches uniform sampling accuracy with 5–10× fewer queries.

1 Motivation

Classifying black-box generative models requires querying each model and comparing responses via pairwise energy distances. In a typical workflow, a labeled pilot set of n models is queried on a large pool of M queries, and the cached responses are used to estimate the discriminative structure. The cost of classifying a *new* model scales with the number of queries m it must answer — each query is an API call. Identifying a small, high-quality query set Q^* of size $m \ll M$ that preserves classification accuracy therefore directly reduces the per-model audit cost for all future models.

2 Methods

1. Uniform Sampling (baseline). Sample m queries uniformly at random from the pool of M queries. No pilot information is used.

2. Estimated Signal Sampling. Fit a two-component GMM to the SVD loadings $|U_{q,\ell}|$ of the energy tensor E. Queries assigned to the higher-mean component form the estimated signal set $\hat{\mathcal{S}}$. Sample m queries uniformly from $\hat{\mathcal{S}}$. This exploits zero-set structure but ignores query redundancy.

3. Forward Stepwise Selection. Define a response $z_{ij} = \mathbf{1}[y_i \neq y_j]$ for each model pair and a predictor $E_{q,ij} = \|g(f_i(q)) - g(f_j(q))\|^2$ for each query q. At each step, add the query whose $E_{q,:}$ has the largest partial correlation with the residual of z after projecting out previously selected queries. This is Gram-Schmidt orthogonalization on E w.r.t. z, ensuring each query contributes maximally non-redundant signal. Cost: $O(m \cdot M \cdot \binom{n}{2})$ — negligible.

3 Setup

Task	n	M	Description
Motivating	100	200	Sensitive training data
Sys. Prompt	100	200	Covert persuasion bias
RAG	120	200	Unauthorized doc. stores

$n_{\text{train}} = 80$ models for estimation and classification; rest held out. MDS (8 components) + LDA. Averaged over 100 random splits.

4 Results

Stepwise dominates at small m: at m=2, stepwise error is 10% (motivating) and 11% (system prompt) vs. 28% for uniform. Estimated signal is intermediate, confirming the GMM partition captures real structure. On the system prompt task, stepwise at m=5 matches uniform at m=50 — a 10× reduction in per-model query cost.